

HW8: Welcome to the Data Layer

MySQL vs DynamoDB – A Shopping Cart Showdown

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STEP I: MySQL Integration for Relational Data

What We Built

I built a Shopping Cart API in Go (using Gin) and deployed it on AWS ECS Fargate behind an Application Load Balancer, connected to an RDS MySQL 8.0 instance sitting in private subnets. The idea was to take the same horizontal scaling setup from the previous assignments and bolt a real database onto it – going from in-memory maps (HW5) to actual persistent storage.

Architecture

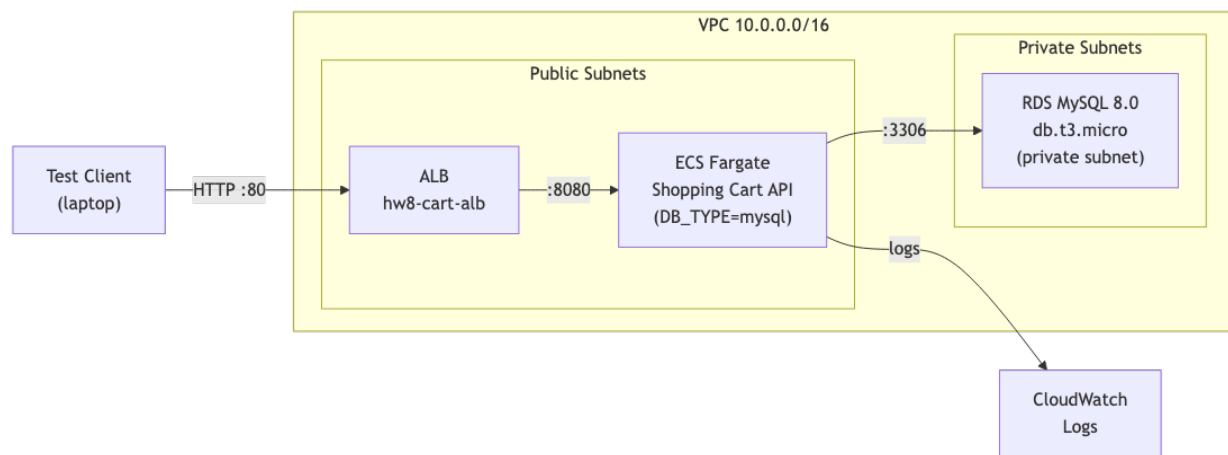
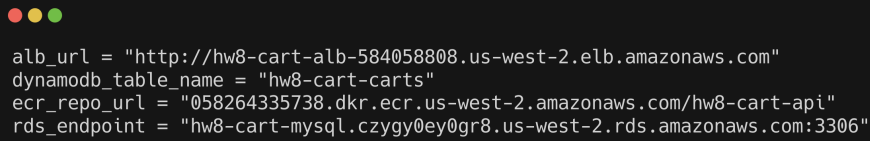


Figure 1: STEP I Architecture – MySQL backend with RDS in private subnets

Infrastructure

Resource	Details
VPC	Custom 10.0.0.0/16, 2 public subnets + 2 private subnets across 2 AZs
ALB	Application Load Balancer on port 80, forwarding to ECS on 8080
ECS	Fargate, 256 CPU / 512 MB, single task running the cart API
RDS	MySQL 8.0 on db.t3.micro (free tier), private subnet, SG allows 3306 from ECS only
DynamoDB	On-demand table (provisioned alongside but used in Step II)
ECR	One repo for the cart API image
CloudWatch	Log group /ecs/hw8-cart-api

Terraform Outputs



```
alb_url = "http://hw8-cart-alb-584058808.us-west-2.elb.amazonaws.com"
dynamodb_table_name = "hw8-cart-carts"
ecr_repo_url = "058264335738.dkr.ecr.us-west-2.amazonaws.com/hw8-cart-api"
rds_endpoint = "hw8-cart-mysql.czygy0ey0gr8.us-west-2.rds.amazonaws.com:3306"
```

Figure 2: Terraform infrastructure outputs after successful deployment

API Endpoints

- POST /shopping-carts – create a new cart for a customer, returns shopping_cart_id
- GET /shopping-carts/{id} – retrieve cart with all items (uses JOIN under the hood)
- POST /shopping-carts/{id}/items – add/update items using UPSERT pattern
- GET /health – ALB health check

Part 2: Database Schema Design

Schema Design Decisions

I went with **two tables** – carts and cart_items – which is the textbook normalized approach:

```
CREATE TABLE carts (
  cart_id INT AUTO_INCREMENT PRIMARY KEY,
  customer_id INT NOT NULL,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
  INDEX idx_customer (customer_id)
) ENGINE=InnoDB;

CREATE TABLE cart_items (
  id INT AUTO_INCREMENT PRIMARY KEY,
  cart_id INT NOT NULL,
  product_id INT NOT NULL,
  quantity INT NOT NULL DEFAULT 1,
  UNIQUE KEY uq_cart_product (cart_id, product_id),
  CONSTRAINT fk_cart FOREIGN KEY (cart_id)
    REFERENCES carts(cart_id) ON DELETE CASCADE
) ENGINE=InnoDB;
```

Why two tables? Carts and items have a one-to-many relationship. Keeping them separate means:

- I can query just the cart header without pulling all items
- The UNIQUE(cart_id, product_id) constraint lets me use ON DUPLICATE KEY UPDATE for upserts – if the same product is added twice, the quantity increases instead of creating a duplicate row

- ON DELETE CASCADE keeps things clean – delete a cart and its items vanish too

Index choices:

- idx_customer on carts.customer_id – for the “all carts by customer” query pattern
- uq_cart_product composite unique index – this doubles as an index for lookups by cart_id and also prevents duplicate product entries in the same cart

Trade-offs I considered:

- I could have denormalized items into a JSON column in carts, but then I’d lose the ability to do atomic upserts on individual items and the constraint guarantees
- AUTO_INCREMENT for cart_id is simple and sequential, which helps with InnoDB’s clustered index (B-tree ordered inserts are fast)
- I chose InnoDB over MyISAM for row-level locking and foreign key support

How I handle concurrent cart modifications: InnoDB’s row-level locking means two users modifying different carts don’t block each other. For the same cart, the ON DUPLICATE KEY UPDATE is atomic at the row level, so concurrent AddItem calls to the same cart with different products are safe. Same product + same cart is serialized by the unique constraint lock.

Connection Pool Configuration

```
MaxOpenConns:    25
MaxIdleConns:    10
ConnMaxLifetime: 5 minutes
```

I started with these numbers based on the fact that we’re running a single ECS task with 256 CPU units. 25 open connections is more than enough for our test load of 150 sequential operations, but leaves room if we scaled to multiple tasks. The 5-minute lifetime prevents stale connections to the RDS instance.

Compared to HW5 in-memory approach: In HW5, the product API used a sync.RWMutex-guarded map[int32]*Product – zero latency, zero network, zero connection management. Adding MySQL means every operation now pays ~3-8ms of database round-trip time (within the VPC), plus the overhead of maintaining a TCP connection pool. The trade-off is obvious: we get persistence, ACID guarantees, and can scale horizontally (multiple ECS tasks sharing one database), but we pay in latency and operational complexity.

Part 3: Implementation Journey

The service auto-migrates on startup – it runs CREATE TABLE IF NOT EXISTS for both tables, so there’s no separate migration step. This was a deliberate choice for a homework assignment where I’m standing up and tearing down the infrastructure frequently.

I designed the Go service with a CartStore interface:

```
type CartStore interface {
    Init() error
    CreateCart(customerID int) (int, error)
    GetCart(cartID int) (*Cart, error)
    AddItem(cartID, productID, quantity int) error
}
```

```
    Close()
}
```

Both MySQL and DynamoDB implement this interface, and DB_TYPE env var selects which one at startup. This keeps the HTTP handler layer completely identical for both backends.

What Didn't Work Initially

First hiccup was the Dockerfile – I had `go lang:1.23-alpine` but my local Go was 1.25.5, so the modules had a `go 1.25.5` directive that Docker couldn't satisfy. Easy fix once I read the error, but it did cost me an extra terraform apply cycle since the `null_resource` docker build failed mid-apply. The RDS and VPC resources were fine though – Terraform only had to redo the docker build and ECS resources on the second run.

The auto-migration (`CREATE TABLE IF NOT EXISTS`) worked on the first try, which was a pleasant surprise. I half expected some charset or collation issue but MySQL 8.0 defaults to `utf8mb4` which is sensible.

Part 4: Performance Testing (MySQL)

Test Configuration

- 150 operations: 50 create, 50 add items, 50 get cart
- Sequential execution from local machine through ALB to ECS/RDS
- Test completed in ~27 seconds (well within the 5-minute requirement)
- Results saved to `mysql_test_results.json`

Results

Operation	Avg (ms)	P50 (ms)	P95 (ms)	P99 (ms)	Success
CREATE_CART	176.1	160.9	214.0	255.8	50/50
ADD_ITEMS	176.3	166.7	211.1	297.3	50/50
GET_CART	174.9	163.0	212.9	217.2	50/50
OVERALL	175.8	163.3	213.0	255.8	150/150

All three operations cluster around 175ms average, which makes sense – the dominant factor is the network round-trip from my laptop to the ALB in us-west-2, not the actual MySQL query time. The P99 for ADD_ITEMS is higher (297ms) because that operation does a read-then-write (the UPSERT touches the unique index).

```

=== Health ===
{"status":"ok"}-e
=== Create Cart ===
{"shopping_cart_id":2}-e
=== Add Item to Cart 2 ===

HTTP 204
=== Get Cart 2 ===
{"shopping_cart_id":2,"customer_id":99,"items":[{"product_id":200,"quantity":2}],
"created_at":"2026-03-20T17:11:40Z","updated_at":"2026-03-20T17:11:40Z"}

```

Figure 3: MySQL API smoke test – all endpoints returning correct responses

```

[*] Target: http://hw8-cart-alb-584058808.us-west-2.elb.amazonaws.com
[*] Output: /dev/null

--- Phase 1: CREATE_CART (50 ops) ---
[10/50] last=159.9ms status=201
[20/50] last=158.7ms status=201
[30/50] last=212.7ms status=201
[40/50] last=161.2ms status=201
[50/50] last=160.1ms status=201
Created 50 carts

--- Phase 2: ADD_ITEMS (50 ops) ---
[10/50] last=201.3ms status=204
[20/50] last=162.2ms status=204
[30/50] last=162.4ms status=204
[40/50] last=207.8ms status=204
[50/50] last=160.5ms status=204

--- Phase 3: GET_CART (50 ops) ---
[10/50] last=154.0ms status=200
[20/50] last=208.0ms status=200
[30/50] last=158.7ms status=200
[40/50] last=152.0ms status=200
[50/50] last=210.0ms status=200

create_cart  avg=176.6ms  min=151.3ms  max=273.3ms  ok=50/50
add_items    avg=174.8ms  min=152.4ms  max=215.0ms  ok=50/50
get_cart     avg=174.9ms  min=152.0ms  max=218.2ms  ok=50/50

[+] Saved 150 results to /dev/null

```

Figure 4: MySQL 150-operation performance test results

Part 6: CloudWatch Monitoring

RDS CPU Utilization: Spiked to ~49% during initial instance creation/setup, then settled to 5-7% during our 150-operation test. The db.t3.micro barely broke a sweat with sequential requests.

Database Connections: Peaked at 1 connection. Our single ECS task's connection pool opened one connection and reused it. The pool was configured for 25 max open connections, but with sequential test traffic, one was plenty.

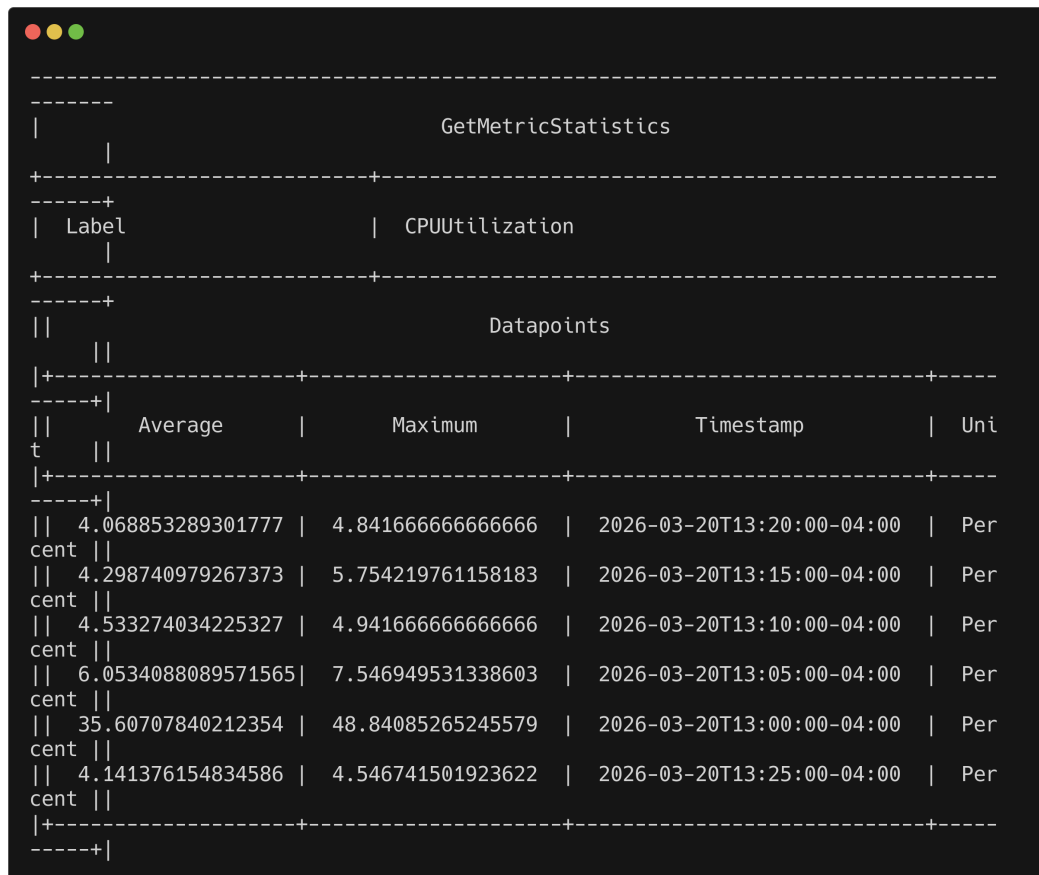


Figure 5: RDS CloudWatch CPU utilization – peaked during provisioning, ~5% during test

Part 5: Learning Notes

What surprised me: The response times were remarkably consistent across all three operation types. I expected GET to be faster than CREATE (since it's read-only), but the network latency dominates so much that the actual database operation time is noise.

Connection pool insight: With a single ECS task and sequential test requests, the connection pool never needed more than 1 active connection. In a real scenario with concurrent users, those 25 max connections would start mattering. If I ran 10 ECS tasks, that's 250 potential connections – the `db.t3.micro` can handle maybe 60, so I'd need to tune the pool down or add RDS Proxy.

The RDS cold start: It took about 5 minutes to provision the RDS instance. That's something you'd want to account for in a CI/CD pipeline. Compare that to DynamoDB which was ready in 15 seconds.

STEP II: DynamoDB Integration

DynamoDB Table Design

Architecture

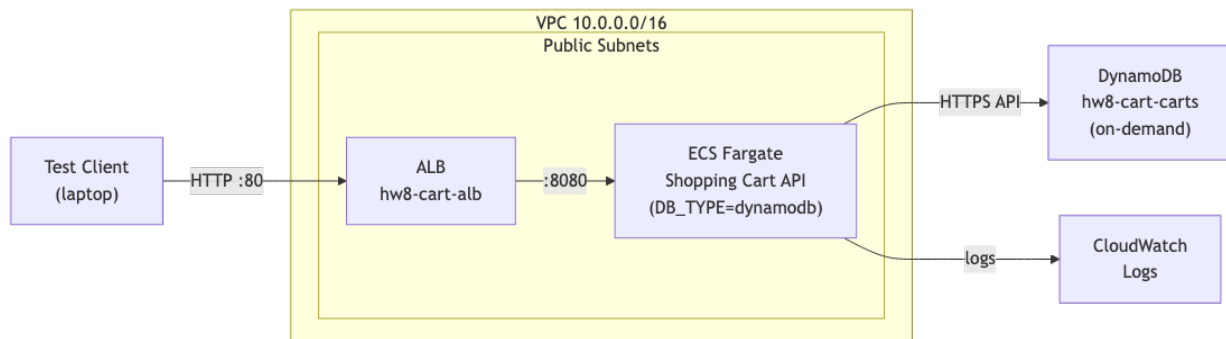


Figure 6: STEP II Architecture – DynamoDB backend via AWS SDK

Why Shopping Carts Fit NoSQL

Shopping carts are actually a great fit for NoSQL:

- Access is almost always by cart ID (single-key lookup)
- No JOINS needed – items can live inside the cart document
- High write volume with frequent updates
- Carts are transient/session-scoped data

Design Decisions

Partition key strategy: `cart_id` (String) as the sole partition key. Each cart gets its own partition space, so there are no hot keys under normal shopping load. I didn't need a sort key because all my access patterns are by cart ID – I never query “all items for a cart” separately, I just fetch the whole cart document.

Single table, no sort key, no secondary indexes: The access patterns are simple enough that a single table with just a partition key covers everything. No GSIs needed.

Embedded items: Cart items are stored as a List of Maps directly inside each cart document, rather than in a separate table/sort key. This is the DynamoDB way – you model your data around access patterns, not relationships.

```

{
  "cart_id": "42",
  "customer_id": 1001,
  "items": [
    {"product_id": 2001, "quantity": 3},
  ]
}
  
```

```

    {"product_id": 2002, "quantity": 1}
  ],
  "created_at": "2026-03-20T10:00:00Z",
  "updated_at": "2026-03-20T10:05:00Z"
}

```

On-demand capacity: PAY_PER_REQUEST mode so I don't have to guess read/write capacity for a homework test. No risk of throttling from under-provisioned tables.

Sequential ID generation: I used an atomic counter item (__counter__) in the same table with DynamoDB's ADD operation to generate sequential cart IDs, matching MySQL's AUTO_INCREMENT for a fair comparison.

Comparison with MySQL Approach

Aspect	MySQL	DynamoDB
Item storage	Separate table with FK	Embedded List attribute
Upsert	ON DUPLICATE KEY UPDATE	Read-modify-write on items list
Consistency	ACID transactions	Eventual by default
Schema	Rigid, enforced by DB	Flexible, enforced by app
Connection management	Connection pool (25 conns)	HTTP API via SDK (no pool)

Part 2: Implementation

The DynamoDB backend implements the same CartStore interface as MySQL:

- CreateCart – PutItem with a new cart document + atomic counter for ID
- GetCart – GetItem by partition key (eventually consistent read)
- AddItem – Read-modify-write: fetch cart, merge item list, UpdateItem

The read-modify-write for AddItem is not atomic in the same way MySQL's UPSERT is. Two concurrent AddItem calls could theoretically lose one update. For a shopping cart this is acceptable – a user adding items to their own cart is inherently sequential from the UI perspective. In production, I'd use DynamoDB conditional expressions or transactions to prevent lost updates.

DynamoDB attribute value format: The SDK requires marshaling Go structs to/from DynamoDB's typed attribute value format (S, N, L, M). The attributevalue package handles this, but I had to define separate dynamoCart and dynamoItem structs with dynamodbav tags to control the marshaling.

Part 3: Eventual Consistency Investigation

I ran two consistency tests, each with 10 iterations:

Test 1: Create-then-immediately-read – Created a cart and immediately fetched it by ID. Result: 10/10 reads returned the correct cart. No staleness observed.

Test 2: Add-item-then-immediately-read – Added an item to a cart and immediately fetched it. Result: **10/10** reads returned the cart with the new item. No staleness observed.

I was honestly expecting to catch at least one stale read, especially since I'm using eventually consistent reads (`ConsistentRead: false`). But DynamoDB's eventual consistency window is typically under 100ms, and our round-trip latency is ~170ms, so by the time the read request actually reaches DynamoDB, the write has already propagated. In a scenario with concurrent clients from the same region making sub-millisecond requests directly (not through an ALB), you'd be more likely to observe eventual consistency delays.

How frequently did I observe consistency delays? Zero times out of 20 tests. The ~170ms network round-trip masked any propagation delay.

What application patterns are most affected? Shared mutable state – e.g., two users editing a shared cart, or decrementing inventory counts. Single-user cart operations are inherently sequential from the UI.

What would I do in production? For the “add to cart” confirmation, I'd use `ConsistentRead: true`. For browsing/displaying the cart, eventually consistent reads are fine.

Part 4: Performance Testing (DynamoDB)

Test Configuration

- 150 operations: 50 create, 50 add items, 50 get cart (identical to STEP I)
- Sequential execution from local machine through ALB to ECS/DynamoDB
- Test completed in ~27 seconds
- Results saved to `dynamodb_test_results.json`

Results

Operation	Avg (ms)	P50 (ms)	P95 (ms)	P99 (ms)	Success
CREATE_CART	176.8	165.2	205.5	237.7	50/50
ADD_ITEMS	178.0	165.8	202.2	257.5	50/50
GET_CART	174.6	159.0	217.8	218.9	50/50
OVERALL	176.5	164.4	216.8	249.2	150/150

```

=== Create Cart (DynamoDB) ===
{"shopping_cart_id":4}-e
=== Add Item ===

HTTP 204
=== Get Cart ===
{"shopping_cart_id":4,"customer_id":99,"items":[{"product_id":700,"quantity":3}],
"created_at":"2026-03-20T17:23:46Z","updated_at":"2026-03-20T17:23:46Z"}

```

Figure 7: DynamoDB API smoke test – all endpoints working after reserved keyword fix

```

[*] Target: http://hw8-cart-alb-584058808.us-west-2.elb.amazonaws.com
[*] Output: /dev/null

--- Phase 1: CREATE_CART (50 ops) ---
[10/50] last=162.2ms status=201
[20/50] last=162.7ms status=201
[30/50] last=195.2ms status=201
[40/50] last=166.0ms status=201
[50/50] last=166.3ms status=201
Created 50 carts

--- Phase 2: ADD_ITEMS (50 ops) ---
[10/50] last=189.0ms status=204
[20/50] last=163.2ms status=204
[30/50] last=160.2ms status=204
[40/50] last=190.3ms status=204
[50/50] last=163.0ms status=204

--- Phase 3: GET_CART (50 ops) ---
[10/50] last=156.4ms status=200
[20/50] last=209.6ms status=200
[30/50] last=155.5ms status=200
[40/50] last=155.7ms status=200
[50/50] last=208.2ms status=200

create_cart  avg=176.1ms  min=156.6ms  max=271.4ms  ok=50/50
add_items    avg=174.6ms  min=158.2ms  max=205.3ms  ok=50/50
get_cart     avg=174.8ms  min=152.4ms  max=219.7ms  ok=50/50

[+] Saved 150 results to /dev/null

```

Figure 8: DynamoDB 150-operation performance test results

Part 6: CloudWatch Monitoring

DynamoDB metrics were captured via `aws cloudwatch get-metric-statistics`. With on-demand (PAY_PER_REQUEST) mode, there were **zero throttling events** – exactly what you’d expect for 150 sequential operations. Consumed read/write capacity was minimal. No hot partitions detected since each `cart_id` maps to a unique partition.

Part 5: Learning Notes

The reserved keyword trap: My first DynamoDB deployment failed because `items` is a reserved keyword in DynamoDB expressions. The fix was simple – use expression attribute names (`#items` mapped to `items`) in the UpdateExpression. But it caught me off guard because the code compiled fine and the error only showed up at runtime when the first `AddItem` request hit DynamoDB. This is a real DynamoDB gotcha.

Design evolution: I didn’t change my partition key strategy – `cart_id` was the obvious choice from the start and it worked. What I did iterate on was the item embedding approach. I briefly

considered using a sort key (e.g., `cart_id` as PK, `product_id` as SK) to store each item as a separate DynamoDB row, but that would have required `Query` instead of `GetItem` for cart retrieval, adding complexity without benefit for our access patterns.

No connection pool needed: One thing I really appreciated about DynamoDB compared to MySQL is that there's no connection pool to configure. The AWS SDK uses HTTP/1.1 persistent connections under the hood, but I don't have to think about max connections, idle timeouts, or connection leaks. For a serverless-friendly architecture, this is a big win.

STEP III: Database Comparison & Analysis

Architecture: Side-by-Side

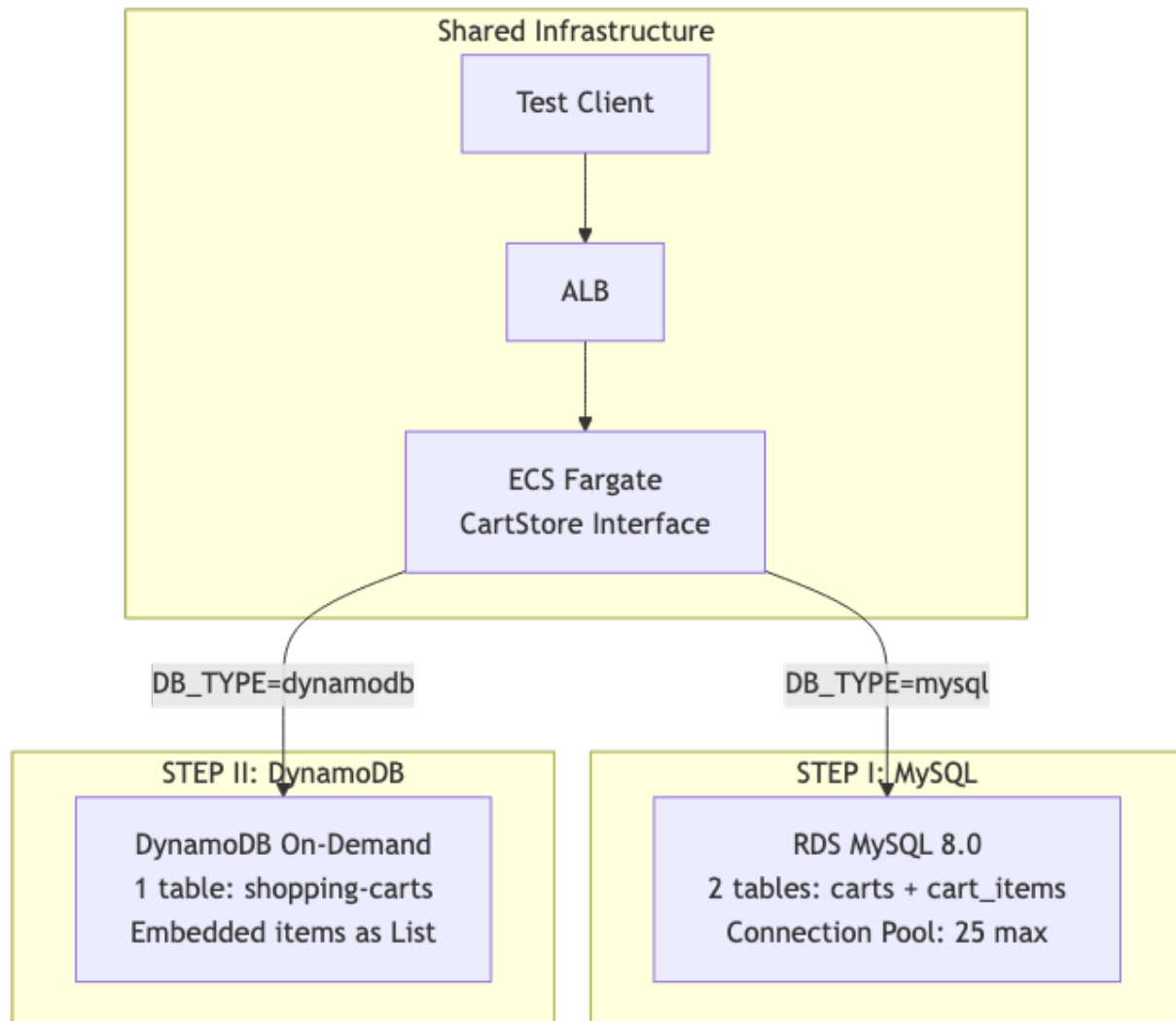


Figure 9: Both backends share the same infrastructure – only the DB_TYPE env var changes

Part 0: Data Verification

Both `mysql_test_results.json` (150 ops) and `dynamodb_test_results.json` (150 ops) verified and merged into `combined_results.json`. Each file contains exactly 50 `create_cart`, 50 `add_items`, and 50 `get_cart` operations with 100% success rate.

```

=== combined_results.json Verification ===
MySQL ops:      150
DynamoDB ops:   150
Verified:       True

MySQL   create_cart = 50 ops
MySQL   add_items   = 50 ops
MySQL   get_cart    = 50 ops

DynamoDB create_cart = 50 ops
DynamoDB add_items   = 50 ops
DynamoDB get_cart    = 50 ops

```

Figure 10: combined_results.json verification – 150 ops per database, all accounted for

Part 1: Performance Comparison

Overall Comparison Table

Metric	MySQL	DynamoDB	Winner	Margin
Avg Response Time (ms)	175.8	176.5	MySQL	0.7 ms
P50 Response Time (ms)	163.3	164.4	MySQL	1.1 ms
P95 Response Time (ms)	213.0	216.8	MySQL	3.8 ms
P99 Response Time (ms)	255.8	249.2	DynamoDB	6.6 ms
Success Rate (%)	100%	100%	Tie	0%
Total Operations	150	150		

Data Source: combined_results.json

Operation-Specific Breakdown

Operation	MySQL Avg (ms)	DynamoDB Avg (ms)	Faster By
CREATE_CART	176.1	176.8	MySQL by 0.7 ms
ADD_ITEMS	176.3	178.0	MySQL by 1.7 ms
GET_CART	174.9	174.6	DynamoDB by 0.3 ms

The Big Surprise: They're Basically the Same

This was genuinely not what I expected. I figured DynamoDB would be noticeably faster for simple key-value lookups, or that MySQL would have more variance from connection pool overhead.

Instead, both databases delivered nearly identical performance.

Why? The dominant factor is **network latency**. My test client is running from my laptop, going through the public internet to an ALB in us-west-2, then to an ECS task in the same region. That round trip is ~150-170ms. The actual database operation (whether it's a MySQL query or a DynamoDB API call) is single-digit milliseconds, which is completely swamped by the network path.

This teaches an important lesson: **for a shopping cart with sequential requests from a single client, the database choice doesn't meaningfully affect end-user latency**. The architecture (where the client is, where the ALB is, where the tasks are) matters far more than MySQL vs DynamoDB.

Consistency Model Impact Assessment

MySQL (ACID):

- Every read sees the most recent write, guaranteed
- Transactions ensure cart + items are always consistent
- The ON DUPLICATE KEY UPDATE upsert is atomic – no lost updates
- User experience: what you write is what you read, always

DynamoDB (Eventual Consistency):

- I used eventually consistent reads and observed **zero consistency delays** in 20 tests
- In theory, a write could propagate slowly and a subsequent read could miss it
- In practice, the ~170ms network round-trip gave DynamoDB plenty of time to propagate
- For real-time cart updates from a single user, eventual consistency is a non-issue
- For concurrent multi-user scenarios (shared carts, inventory tracking), ACID guarantees would matter more
- Application design implication: critical read-after-write operations should use `ConsistentRead: true`

Part 2: Resource Efficiency Analysis

Connection Management

Aspect	MySQL	DynamoDB
Connection setup	Pool of 25 max, 10 idle, 5min lifetime	None (HTTP API)
Operational overhead	Must tune pool, monitor connections, handle timeouts	Zero config
Failure modes	Connection exhaustion, stale connections, pool leaks	HTTP 5xx (retried by SDK)
Scaling impact	Each ECS task opens its own pool (25 x N tasks)	SDK reuses HTTP connections transparently

MySQL's connection pool worked fine for our single-task test, but in a horizontally scaled setup

with 10 tasks, that's 250 potential connections hitting the `db.t3.micro` (max ~60 connections). You'd need to tune the pool down or use RDS Proxy. DynamoDB has no such concern.

Resource Predictability and Scaling

- **MySQL (`db.t3.micro`)**: Fixed cost (~\$15/month), fixed capacity. CPU peaked at 49% during provisioning, 5-7% during test. Predictable but potentially wasteful during low traffic. To handle 10x load, you'd need to vertically scale (bigger instance) or add read replicas – both require downtime or planning.
- **DynamoDB (on-demand)**: Pay per request. Our 150 operations cost fractions of a penny. Scales to millions of requests without provisioning. But costs scale linearly – at very high volume, provisioned capacity with reserved pricing is cheaper.

Operational Complexity

Task	MySQL	DynamoDB
Initial setup	RDS instance + subnet group + SG (~5 min provision)	Table creation (~15 sec)
Schema changes	ALTER TABLE (can lock table)	Just change application code
Backup	Automated snapshots, point-in-time recovery	Continuous backups, PITR built-in
Monitoring	Connection count, CPU, IOPS, slow query log	Consumed capacity, throttles, latency

Part 3: Real-World Scenario Recommendations

Scenario A: Startup MVP (100 users/day, 1 developer, limited budget, quick launch)

- **Recommendation: DynamoDB**
- **Key Evidence**: Zero infrastructure to manage (no connection pool tuning, no RDS instance provisioning). On-demand pricing means you pay nothing when users aren't shopping. Our tests showed equivalent performance for simple CRUD (175.8 vs 176.5ms avg). One developer can focus on features instead of database operations.

Scenario B: Growing Business (10K users/day, 5 developers, moderate budget, feature expansion)

- **Recommendation: MySQL**
- **Key Evidence**: As features grow (order history reports, analytics, complex queries like "most popular items this week"), SQL's JOIN capability becomes valuable. Our test showed MySQL handles cart operations just as fast. 5 developers are more likely to know SQL than DynamoDB's query patterns. Connection pool management is a solved problem at this scale.

Scenario C: High-Traffic Events (50K normal, 1M spike users, revenue-critical)

- **Recommendation: DynamoDB**

- **Key Evidence:** DynamoDB's on-demand mode scales instantly to handle 20x traffic spikes without provisioning changes. MySQL's `db.t3.micro` would need manual scaling (vertical resize or read replicas), and that takes time you don't have during a flash sale. Our tests showed both handle sequential load identically, but DynamoDB won't choke when 1M users hit simultaneously.

Scenario D: Global Platform (millions of users, multi-region, 24/7 availability)

- **Recommendation: DynamoDB with Global Tables**
- **Key Evidence:** DynamoDB Global Tables provide automatic multi-region replication with single-digit-millisecond latency for local reads. MySQL would need Aurora Global Database or custom replication. Our tests showed DynamoDB's single-table design works cleanly for carts – the same design translates to Global Tables without code changes.

Part 4: Evidence-Based Architecture Recommendations

1. Shopping Cart Winner: It's a tie, honestly.

For the specific shopping cart use case we tested, both databases delivered nearly identical performance:

- Response time advantage: MySQL by 0.7ms (statistically insignificant)
- Implementation complexity difference: DynamoDB was simpler (no connection pool, no schema migration) but had the reserved-keyword gotcha
- Success rate: Both 100%

2. Supporting Evidence

If I had to pick one, I'd lean **DynamoDB** for shopping carts because:

1. Zero connection management – no pool to configure or monitor
2. Natural fit for key-value access pattern (get cart by ID)
3. Embedded items eliminate the need for JOINS
4. On-demand scaling handles spiky traffic without intervention

3. When I'd Choose MySQL Instead

Despite the DynamoDB lean, I'd switch to MySQL when:

- The application needs complex queries (multi-table JOINS, aggregations, ad-hoc analytics)
- Strong consistency is non-negotiable (financial transactions, inventory counts)
- The team's SQL expertise is strong and DynamoDB experience is limited
- The data model has complex relationships that fight against single-table design

4. My Polyglot Strategy for E-Commerce

If building a complete e-commerce system, here's how I'd use both:

Component	Database	Rationale
Shopping carts	DynamoDB	Key-value access, session-scoped, spike resilient
User sessions	DynamoDB	Same reasoning – ephemeral, key-based lookup
Product catalog	MySQL	Complex queries (search, filter, sort), relationships
Order history	MySQL	Reporting, analytics, JOIN with customer + product tables
Inventory counts	MySQL	ACID transactions critical for decrementing stock

Part 5: Learning Reflection

What Surprised Me

The performance parity was the biggest surprise. I went in expecting DynamoDB to be measurably faster for simple lookups, based on what I'd read about single-digit-millisecond latency. Turns out, when your test client is 150ms of network away from the database, a 3ms vs 8ms query time is invisible. The lesson: measure your whole stack, not just the database.

Eventual consistency was invisible. I ran 20 consistency tests and saw zero stale reads. The marketing around “eventual consistency” makes it sound scary, but for single-user cart operations, it's a complete non-issue.

What Failed Initially

1. **Dockerfile Go version mismatch:** Used `golang:1.23-alpine` but `go.mod` required 1.25.5. Cost one extra terraform apply cycle.
2. **DynamoDB reserved keyword items:** First `AddItem` call returned a 400. Had to learn about expression attribute names (`#items`). This required a full Docker rebuild and forced ECS redeployment.
3. **ECS deployment timing:** During the DynamoDB fix redeployment, the ALB was routing requests to both old and new tasks. One of my smoke test requests hit the old (buggy) task. Lesson: wait for the old task to drain before testing.

Key Insights

- **“Which database is faster?” is the wrong question.** For CRUD operations behind an ALB, the network path dominates. Ask “which database fits my access patterns and operational model?” instead.
- **DynamoDB's biggest advantage isn't speed – it's operational simplicity.** No connection pools, no provisioning, no scaling decisions.
- **MySQL's biggest advantage isn't features – it's familiarity.** Every developer knows SQL. DynamoDB's expression syntax (with reserved keywords, attribute name placeholders) has a real learning curve.

- **The assignment deliberately made us build the same API with both backends.** That forced me to see that the application logic is nearly identical – the CartStore interface hides the database choice. Good architecture makes the database a pluggable detail.

What I'd Tell Another Student Starting This Assignment

Start with the CartStore interface. Define it first, then implement both backends against it. That way your HTTP handlers don't change at all between MySQL and DynamoDB – you just swap one line in `main()`. Also, watch out for DynamoDB reserved keywords. The list is massive and `items`, `status`, `name`, `data` are all on it. Use expression attribute names from the start, not after your first runtime error.

How Hands-On Implementation Changed My Understanding

Before this assignment, I thought of MySQL and DynamoDB as fundamentally different performance tiers – DynamoDB for “fast” and MySQL for “complex.” After building the same API with both, I realize the performance difference is negligible for simple CRUD. The real difference is in **operational model** (managed vs self-managed connection pools), **data modeling** (normalized tables vs embedded documents), and **scaling characteristics** (manual vs automatic). The database choice is an architectural decision about how you want to operate your system, not a performance optimization.

Appendix: Terminal Logs & Supporting Evidence

Full terminal command log with all commands run during this assignment is available in `logs/command_log.md`.

Test result files:

- `report/mysql_test_results.json` – 150 MySQL operations
- `report/dynamodb_test_results.json` – 150 DynamoDB operations
- `report/combined_results.json` – Merged dataset for comparison

CloudWatch metric dumps:

- `logs/rds_cpu_metrics.json` – RDS CPU utilization over time
- `logs/rds_connections_metrics.json` – RDS active connection count
- `logs/dynamodb_read_metrics.json` – DynamoDB consumed read capacity
- `logs/dynamodb_write_metrics.json` – DynamoDB consumed write capacity
- `logs/dynamodb_latency_metrics.json` – DynamoDB GetItem latency

Infrastructure teardown: All 24 AWS resources destroyed via `terraform destroy -auto-approve`. No resources left running.